

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Sowmiya. D (192511063) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sowmiya. D

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Article Review: Machine Learning for Gasoline Recognition in Fire Debris

Selected Article

Bogdal, C., Schellenberg, R., Höpli, O., Bovens, M., & Lory, M. (2022). "Recognition of gasoline in fire debris using machine learning: Part I, application of random forest, gradient boosting, support vector machine, and naïve Bayes." *Forensic Science International*, 331, 111146. DOI: 10.1016/j.forsciint.2021.111146. The article was published online in December 2021 and appeared in the February 2022 issue. ([PubMed](#))

DOI: <https://doi.org/10.1016/j.forsciint.2021.111146>

Publisher/Article page: [Elsevier ScienceDirect article page](#)

Domain and ML problem

The application domain is forensic science, specifically fire/arson investigation. The ML problem is a binary supervised classification problem: determine whether a fire-debris sample contains gasoline residue ("gasoline") or does not contain gasoline ("no gasoline") from gas chromatography–mass spectrometry (GC-MS) measurements. The authors compare Random Forest (RF), Gradient Boosting Machine (GBM), Support Vector Machine (SVM), and Naïve Bayes (NB). Logistic Regression was also attempted but did not converge because the classes were strongly separated. ([ScienceDirect](#))

Task 1: Article Summary

1(a) Citation, domain and problem

Full citation:

Bogdal, C., Schellenberg, R., Höpli, O., Bovens, M., & Lory, M. (2022). Recognition of gasoline in fire debris using machine learning: Part I, application of random forest, gradient boosting, support vector machine, and naïve Bayes. *Forensic Science International*, 331, 111146. <https://doi.org/10.1016/j.forsciint.2021.111146>. ([PubMed](#))

The study belongs to the forensic chemistry/forensic fire investigation domain. Traditional fire-debris analysis requires a forensic examiner to interpret complicated GC-MS results. Because fire debris contains pyrolysis products and other chemical interferences, visual interpretation can be difficult and time-consuming. The ML task is therefore to automate or assist the preliminary recognition of gasoline residues. ([ScienceDirect](#))

1(b) Key objective and research gap

The main objective of the article is to determine whether classical machine-learning algorithms can reliably recognize gasoline residues in fire-debris samples from GC-MS data. In conventional forensic practice, the analyst must inspect complex chromatographic and mass-spectrometric patterns and decide whether an ignitable liquid is present. This process can be difficult because fire debris contains numerous compounds originating from both the original material and combustion/pyrolysis processes. The authors identified a practical gap: although chemometric and AI approaches had already been explored, there was a need for a practical ML-based screening method specifically evaluated using gasoline-containing fire debris, including samples resembling real forensic casework. ([ScienceDirect](#))

To address this gap, the researchers constructed a database of 360 GC-MS measurements containing real-case fire debris and experimentally prepared samples containing weathered gasoline. They trained and compared Random Forest, Gradient Boosting, Support Vector Machine, and Naïve Bayes classifiers. Logistic

Regression was also tested but failed to converge. The results showed that RF, GBM and SVM could correctly classify the test samples, suggesting that ML can provide a rapid preliminary screening tool while leaving final forensic interpretation to an expert. ([ScienceDirect](#))

Task 2: Methodology Analysis

2(a) ML techniques and dataset

Machine-learning techniques

The authors evaluated four main supervised classification techniques:

1. Random Forest (RF)

Random Forest is an ensemble of decision trees. Multiple trees are trained using different subsets/features of the training data, and their predictions are combined. This makes RF relatively robust and also allows estimation of feature importance. In this study, RF was particularly useful because it could identify chemical compounds that contributed strongly to gasoline classification. ([ScienceDirect](#))

2. Gradient Boosting Machine (GBM)

GBM constructs a sequence of weak decision-tree models, where each subsequent tree attempts to correct errors made by previous trees. It is suitable for complex nonlinear relationships in high-dimensional data.

3. Support Vector Machine (SVM)

SVM is a statistical learning method that attempts to find an effective separating boundary between classes. It is particularly appropriate when the feature space is high-dimensional. Here, the two classes were “with gasoline” and “without gasoline.”

4. Naïve Bayes (NB)

Naïve Bayes is a probabilistic statistical classifier based on Bayes' theorem. It assumes conditional independence between features given the class. The study demonstrated an important weakness of NB: it had difficulty recognizing gasoline-like oils/petrol that were not represented in the training data. ([ScienceDirect](#))

5. Logistic Regression

Logistic Regression was also attempted. However, it failed to converge because the gasoline and non-gasoline classes were too well separated. Therefore, it was excluded from subsequent evaluation. ([ScienceDirect](#))

Dataset

The dataset contained 360 GC-MS measurements. It included:

- Fire-debris samples from real cases.
- Fire-debris samples experimentally spiked with known quantities of weathered gasoline.
- Gasoline samples weathered to different degrees.
- Weathered gasoline levels reaching approximately 99.6% in the experimental weathering procedure.
- Fire-debris samples without gasoline to provide the negative class. ([ResearchGate](#))

The supplementary material indicates that gasoline was collected from different sources, including fuel stations and fuel tanks. Fourteen gasoline samples were collected, with nine subsequently used in weathering

experiments. Fire debris free of ignitable liquid was then spiked with approximately 0.5–20 μL of weathered gasoline. ([ResearchGate](#))

The GC-MS measurements were processed to obtain comparable analytical variables. Retention-time variation was controlled during chromatographic processing; a later review of the work notes that retention-time locking was performed using compounds including toluene/benzene at an early stage and naphthalene at a later stage. ([ScienceDirect](#))

The data were divided into approximately 85% training data and 15% test/validation data. Thus, the experiment was designed to evaluate how well the trained classifiers could recognize previously unseen samples. ([ScienceDirect](#))

2(b) Mapping to CO4 topics

CO4 Topic	Connection to the Article
Inductive Learning	The algorithms learn general classification rules from labelled GC-MS examples and apply those rules to previously unseen fire-debris samples.
Decision Trees	Random Forest and Gradient Boosting are tree-based ensemble methods.
Statistical Learning	SVM and Naïve Bayes are directly related to statistical learning methods specified in CO4.
Reinforcement Learning	Not used. The problem is supervised classification rather than sequential decision-making.

Methodological strength

A major strength is the comparison of multiple fundamentally different classifiers on the same forensic problem. RF, GBM, SVM and NB provide different modelling assumptions, allowing the researchers to determine whether the observed performance is dependent on a particular algorithm. The RF model also provides information about important chemical compounds, improving interpretability. ([ScienceDirect](#))

Methodological limitation

The principal limitation is the relatively small and narrowly defined dataset. Although 360 measurements sounds substantial, they represent a highly specific problem: gasoline recognition rather than recognition across the full range of ignitable liquids. The authors themselves acknowledge that the dataset is limited and that successful error-free classification depends on the machine being trained with samples similar to those encountered during testing. ([ScienceDirect](#))

Task 3: Results and Critical Evaluation

3(a) Key results

The authors reported very strong classification performance for three of the four successful models.

Model	Result reported by authors	Interpretation
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Random Forest	100% correct classification of test samples	No false-positive or false-negative allocation reported for the test set
Gradient Boosting Machine	100% correct classification of test samples	No false-positive or false-negative allocation reported
Support Vector Machine	100% correct classification of test samples	No false-positive or false-negative allocation reported
Naïve Bayes	Performed well on trained classes but had difficulty with gasoline-like oils/petrol absent from training data	Generalization problem
Logistic Regression	Did not converge	Excluded from further evaluation

The central finding was that RF, GBM and SVM classified all samples in the test/validation set correctly, whereas Naïve Bayes struggled with non-gasoline ignitable liquids that were not represented adequately in the training data. ([ScienceDirect](#))

The supplementary material also examined how the random train/test split affected false-negative and false-positive error rates. Different random seeds produced different estimated error rates, indicating that the exact partition of the data could influence performance estimates. ([ResearchGate](#))

It is important not to invent F1-score or AUC values here: the main article's reported headline result is classification correctness/error allocation rather than a complete set of F1/AUC metrics.

3(b) Critical evaluation

Are the reported metrics realistic?

The reported 100% classification result is possible, but it should not automatically be interpreted as evidence of 100% real-world accuracy.

The strongest reason for caution is dataset size and composition. The dataset contained only 360 measurements, with 15% reserved for testing. The data also included experimentally prepared samples and gasoline samples from a limited set of sources. Therefore, the test distribution may be relatively similar to the training distribution. ([ScienceDirect](#))

Another issue is that the target problem was relatively specific: gasoline versus no gasoline. Real forensic laboratories encounter many other ignitable liquids and complex mixtures. The study itself demonstrated that Naïve Bayes had trouble when exposed to gasoline-like oils and petrol that were not represented in the training data. This is strong evidence that generalization beyond the training distribution remains an important challenge. ([ScienceDirect](#))

Potential biases

1. Dataset/source bias:

The gasoline samples came from a limited number of sources. ML models may unintentionally learn characteristics associated with those particular samples rather than gasoline itself.

2. Laboratory-condition bias:

Experimentally prepared spiked fire debris may not fully reproduce the complexity of actual fire scenes.

3. Train/test similarity:

If chemically similar samples from the same source are distributed between training and test sets, performance may be optimistic.

4. Class/generalization bias:

The study concentrates heavily on gasoline. Other ignitable liquids and unusual mixtures are not adequately represented.

5. Limited independent validation:

A stronger evaluation would involve a completely independent dataset collected by another laboratory using different instruments, operators, gasoline brands and substrates.

Additional experiments that would strengthen the research

The study could be strengthened through:

1. External validation using a dataset from another forensic laboratory.
2. Leave-one-source-out validation, where gasoline from one station/source is completely excluded from training and used only for testing.
3. Cross-laboratory validation using different GC-MS instruments.
4. Testing on a much larger dataset containing gasoline, diesel, kerosene and other ignitable liquids.
5. Reporting precision, recall, F1-score, specificity and ROC-AUC, rather than relying mainly on classification correctness.
6. Testing realistic mixtures containing gasoline plus multiple pyrolysis products.
7. Performing repeated stratified cross-validation and reporting confidence intervals.

These experiments would establish whether the apparent 100% performance represents genuine generalization or mainly successful recognition of a relatively controlled dataset.

3(c) Real-world applicability

The proposed system could be deployed as a screening tool in forensic laboratories. A fire-debris sample could first be analysed using GC-MS, after which the ML classifier could rapidly indicate whether the chemical pattern is consistent with gasoline.

Potential applications include:

- Preliminary screening of suspected arson samples.
- Prioritizing samples for detailed forensic examination.
- Supporting forensic examiners when large numbers of samples must be processed.
- Providing an additional independent computational opinion.
- Eventually extending the approach to other ignitable liquids.

The authors emphasize that the system should support rather than replace the forensic examiner. Final identification of an ignitable liquid should still be performed by a qualified expert. ([ScienceDirect](#))

Challenges in industry

Data availability: Large, diverse and properly labelled forensic datasets are difficult to obtain because real casework is sensitive and legally controlled.

Instrumentation differences: GC-MS instruments, columns, operating conditions and laboratories can differ. A model trained on one laboratory's data may lose accuracy when deployed elsewhere.

Computational requirements: Classical models such as NB, SVM and RF are comparatively practical, so computational cost is less problematic than with deep neural networks. However, preprocessing and standardized GC-MS data acquisition remain important.

Ethical/legal issues: In forensic applications, an incorrect prediction could influence an investigation or judicial decision. Therefore, model validation, explainability, uncertainty estimation and human oversight are essential.

Model drift: Fuel formulations and laboratory procedures can change over time. The model would need periodic validation and retraining.

Task 4: Reflection and Learning Outcome

4(a) Three key insights

Insight 1 — Supervised learning is fundamentally dependent on representative training data

The most important lesson is that high classification accuracy does not automatically mean strong generalization. Naïve Bayes performed poorly when confronted with gasoline-like liquids that were not adequately represented in its training data. This connects directly to CO4: Inductive Learning. An inductive learner derives a general rule from examples, so the quality and diversity of those examples determine how well the learned rule transfers to unseen cases.

Insight 2 — Different statistical models make different assumptions

The comparison between SVM and Naïve Bayes demonstrates that statistical learning algorithms are not interchangeable. Naïve Bayes relies on conditional-independence assumptions, while SVM seeks a discriminating decision boundary. The different behaviour of the models illustrates the CO4 Statistical Learning concept that algorithm selection should depend on the structure of the data and the application.

Insight 3 — Ensemble tree methods can provide both prediction and interpretation

Random Forest and Gradient Boosting achieved excellent classification performance, while Random Forest additionally helped identify chemically important variables. This connects to CO4 Decision Trees and shows why tree-based ensemble methods are useful: combining many trees can improve predictive performance while feature importance can provide some insight into the underlying decision process.

4(b) Proposed extension/novel experiment

I would extend the research by developing a leave-one-laboratory-and-source-out evaluation using a multi-ignitable-liquid dataset.

The experiment would collect GC-MS data from several independent forensic laboratories and include gasoline, diesel, kerosene, lighter fluid and other common ignitable liquids. Instead of randomly splitting individual measurements, all samples originating from one laboratory or gasoline source would be completely excluded from training and used as an external test set.

The experiment could compare the original SVM, Naïve Bayes, Random Forest and Gradient Boosting models with a modern calibrated ensemble model. Performance would be measured using accuracy, precision, recall, F1-score, specificity, ROC-AUC and confidence intervals.

This enhancement is feasible because the original study already establishes the GC-MS-to-ML workflow. Its expected impact would be substantial: it would determine whether the excellent reported performance survives a much harder distribution-shift test. If performance remains high, confidence in forensic deployment would increase considerably. If performance decreases, the experiment would identify the need for domain adaptation, calibration or additional training data before operational use.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand